Predicting the Earth Similarity Index of Exoplanets Using Stellar and Orbital Features

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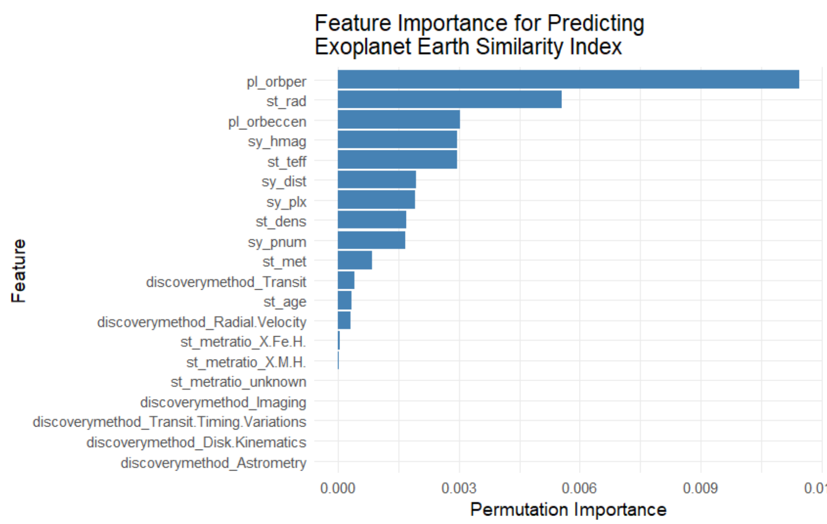
Abstract

This paper investigates whether the Earth Similarity Index (ESI) of a confirmed exoplanet can be predicted using only stellar and orbital features, without any knowledge of a planet's physical composition. Data from NASA Exoplanet Archive's Planetary Systems Composite dataset and the Habitable Worlds Catalog (HWC) from the PHL @ UPR Arecibo were combined and used to train a random forest regression model. The results suggest that stellar and orbital features carry meaningful, but incomplete, information for predicting Earth-likeness. Feature importance analysis identified orbital period, stellar radius, and orbital eccentricity as the most important predictors.



1. Introduction

The NASA Exoplanet Archive currently catalogs over 6,000 confirmed exoplanets (NASA Exoplanet Archive, 2026), and a key point of curiosity is identifying which of these exoplanets closely resemble Earth. One metric that quantifies how Earth-like a planet is is the Earth Similarity Index (ESI), which is provided by the Habitable Worlds Catalog (HWC) maintained by the Planetary Habitability Laboratory (PHL) @ UPR Arecibo. The ESI is a continuous score from 0 to 1 where 0 indicates that a planet is completely different from Earth and 1 indicates that a planet is identical to Earth (Planetary Habitability Laboratory, 2024).

Jiang et al. (2024) combined data from NASA Exoplanet Archive's Planetary Systems Composite dataset with the HWC data and trained a Random Forest classifier to predict the habitability of exoplanets. They conducted a feature importance analysis on the Random Forest classifier, which identified stellar radius, stellar effective temperature, and stellar surface gravity as the most influential predictors of habitability. While their approach has high classification accuracy, it treats habitability as a binary variable, which limits the nuance of the analysis.

In this study, we treat Earth-likeness as a continuous measure by using ESI as the response variable. Furthermore, the features that we use are restricted exclusively to stellar and orbital characteristics. We explicitly exclude any features that directly describe a planet's physical composition to test whether we can predict Earth-likeness using only stellar and orbital features, without any knowledge of a planet's physical composition.

2. Methodology

2.1 Data Sources

This study utilizes two publicly available datasets. The primary source is the Planetary Systems Composite dataset from NASA Exoplanet Archive (2026), which provides physical, orbital, and stellar features. This dataset consolidates multiple observations for a given exoplanet into a single row. The second dataset is the Habitable Worlds Catalog (HWC), maintained by the PHL @ UPR Arecibo, which provides ESI scores for confirmed exoplanets (Planetary Habitability Laboratory, 2024). The two datasets were joined on planet name, retaining only planets present in both datasets.

2.2 Data Preparation

Three initial filtering steps were applied to the combined dataset. First, planets with a controversial status were filtered out to ensure data reliability. A variable called pl_controv_flag from the Planetary Systems Composite Data from NASA Exoplanet Archive (2026) indicates whether an exoplanet's confirmation status is currently disputed. Second, to maintain consistency in assessing stellar radiation and orbital mechanics, the analysis was limited to exoplanets orbiting a single known host star, consistent with Jiang et al. (2024). Third, planets with missing ESI values were excluded since ESI is the response variable.

Features that directly describe a planet's physical composition, such as planetary radius, density, mass, and surface temperature, were explicitly excluded from the final dataset. The remaining features fall into three broad categories. Orbital features capture the geometry and dynamics of the planet's orbit, such as orbital period. Stellar features describe the physical properties of the host star, such as stellar temperature. Lastly, system-level features describe broader properties of the planetary system and its observational characteristics, such as distance to the system and discovery method.

The combined dataset was then partitioned using a stratified 80/20 split on ESI to ensure that the distribution of ESI values was comparable across the training and test sets. Figure 1 shows the ESI distributions in the training and test sets. Both sets exhibit a heavy right skew, with most exoplanets having an ESI below 0.4, which reflects the fact that Earth-like planets are rare among confirmed exoplanets. The near-identical shape of the two distributions shows us that stratified random sampling produced comparable splits.

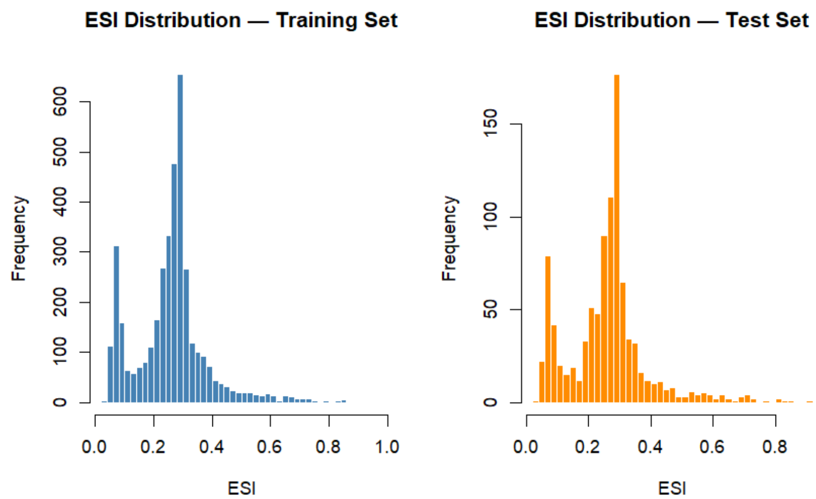


Figure 1. Distribution of Earth Similarity Index (ESI) values in the training set (left, blue) and test set (right, orange) after preprocessing.

2.3 Preprocessing

All preprocessing parameters were derived exclusively from the training set and then applied to the test set, preventing data leakage. The stellar spectral type column (st_spectype) was decomposed into three ordinal features: spectral class, which was encoded from 1 (hottest, O-type) to 7 (coolest M-type); spectral subclass, which was extracted as a numerical value and captures finer temperatures within each spectral class; and luminosity class, which was encoded from 1 (least luminous, white dwarf) to 10 (most luminous, hypergiant). The original st_spectype column was then removed.

Following this, features with more than 25% missing values were removed, which follows the threshold used by Jiang et al. (2024), as were near-zero variance predictors, which offered little to no information. In the discovery method and metallicity ratio columns, missing values were assigned an "unknown" category before being one-hot encoded. Remaining rows with any missing predictor values were then removed. Lastly, highly correlated numeric and ordinal features were identified and removed.

2.4 Model and Evaluation

A Random forest regression model was fitted on the training data. Random forest was chosen for three reasons. First, it captures non-linear relationships without requiring an explicit functional form. Second, it is robust to correlated predictors, as each tree randomly samples a subset of features at each split. Lastly, the model provides feature importance measures, which allows us to identify the stellar and orbital properties that are most predictive of Earth-likeness.

A series of random forest models were fitted with tree counts ranging from 10 to 500 in increments of 10, and the model performance converged, making 500 trees a good choice. The minimum node size parameter was tuned by fitting models with different minimum node sizes on the training set and selecting the value that minimized the out-of-bag (OOB) error.

3. Results

The OOB root mean square error (RMSE) on the training set was 0.0639, and the RMSE on the test set was 0.0721. The model generalizes well to unseen data without overfitting. Given that ESI ranges from 0 to 1 and that the majority of ESI values in the dataset are between 0.05 and 0.40, an RMSE of 0.0721 is moderate, which suggests that stellar and orbital features alone are predictive of Earth-likeness, but still carry incomplete information. Figure 2 shows the predicted versus actual ESI values on the test set, with the red diagonal line indicating perfect accuracy. The model predicts well for planets with low ESI values, which make up most of the planets in the dataset. However, it tends to underpredict for planets with ESI values above 0.5. Figure 3 shows the feature importances from the fitted random forest model. Orbital period (pl_orbper) is the most important predictor of Earth-likeness, followed by stellar radius (st_rad), orbital eccentricity (pl_orbeccen), H-band magnitude (sy_hmag), and stellar effective temperature (st_teff). This shows that a combination of orbital and stellar features carries meaningful information for predicting Earth-likeness.

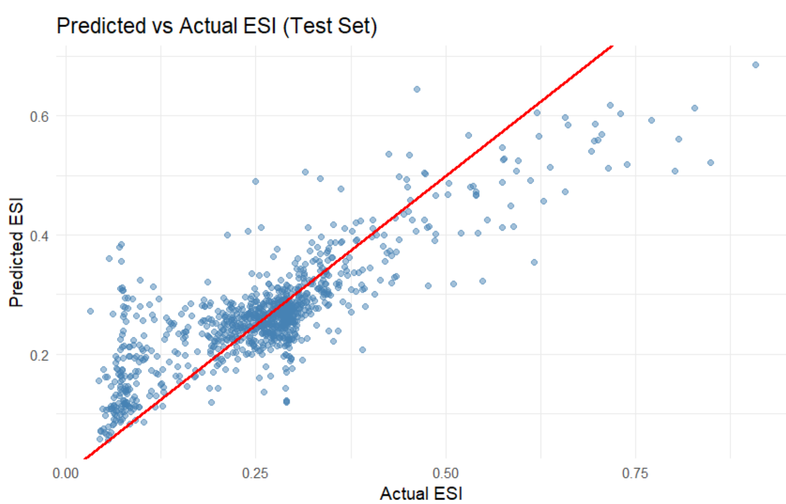


Figure 2. Predicted versus actual ESI values on the test set. The red diagonal line represents perfect prediction.

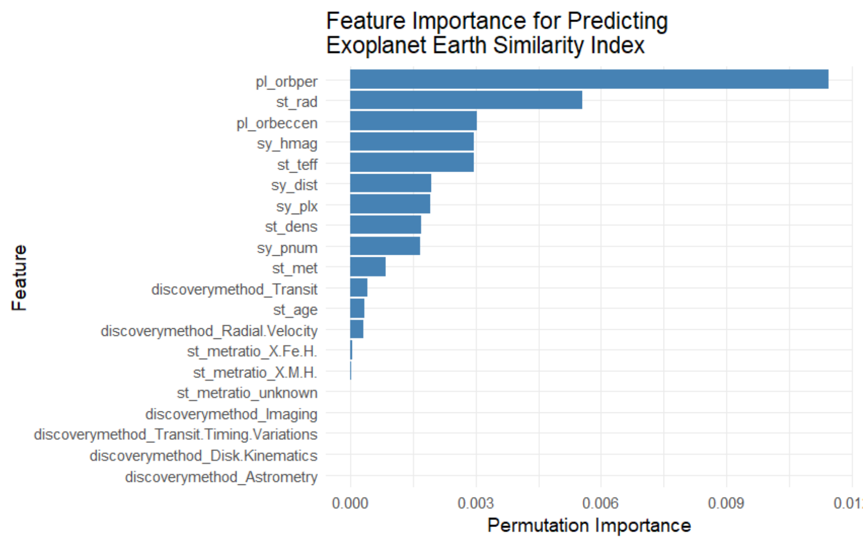


Figure 3. Permutation feature importances for the fitted random forest regression model, sorted in descending order.

4. Conclusion

This study investigated whether the Earth-likeness of a confirmed exoplanet can be predicted using only stellar and orbital features, without any knowledge of a planet's physical composition. The results demonstrate that a Random Forest model with only these features can moderately predict the Earth-likeness (ESI) of an exoplanet, achieving a test RMSE of 0.0721. Orbital period, stellar radius, and orbital eccentricity were identified as the three most important predictors, indicating that a combination of orbital and stellar features carries meaningful, but incomplete, information for predicting Earth-likeness.

References:

1. Jiang, J. H., Rosen, P. E., Liu, C. X., Wen, Q., & Chen, Y. (2024). Analysis of Habitability and Stellar Habitable Zones from Observed Exoplanets. Galaxies, 12(6), 86. https://doi.org/10.3390/galaxies12060086
2. NASA Exoplanet Archive. (2026). Planetary systems composite data. California Institute of Technology. https://exoplanetarchive.ipac.caltech.edu
3. Planetary Habitability Laboratory. (2024). Habitable worlds catalog. University of Puerto Rico at Arecibo. https://phl.upr.edu/hwc